Desktop conversations — the future of multimedia conferencing

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Both the recent explosion of interest in the Internet and the ongoing development of video-on-demand services have focused attention on the retrieval and delivery aspects of telecommunications networks. Yet the telephony network is fundamentally based on a much more basic human need — two-way communication. In fact, people normally communicate using conversation — interactive speech plus additional information in the form of non-verbal cues like body language.

The underlying technologies and standards to enable the easy exchange of information in a conversational way are now converging — which brings together the fields of conferencing, multimedia, and telephony into a unified whole. This could well be the catalyst to turning data communications into the dominant conversational medium of the future.

This paper looks at the ways that the desktop may soon rival the telephone as the natural route for carrying out a media-rich conversation with a remotely located person, and focuses on one example of a desktop conferencing application — Passepartout.

1. Introduction

The history of human beings documents an ongoing desire to communicate at distance. The close-up interaction of body language works at very short distances, while the spoken word can work to large audiences over longer distances. The spoken word is an attempt to translate thoughts into a code which effectively conveys meaning to a listener, and this 'translation and coding' element forms a common thread through all communications theory. Where the spoken word is too quiet, then shouts can be used over longer distances at the cost of a significant loss of both intelligibility and speed of transmission.

Where audio signals fail, then visual methods can be used. Shouting is frequently accompanied by gesticulation — normally arm-waving and exaggerated body movements. The arm waving works much better if the length of the arms and the position of the hands can be emphasized, which results in the use of flags, and ultimately, in telegraph systems using large mechanical arms mounted with lights. The electrical analogue of these telegraph systems shares the same name, and uses simple electrical signals to convey the information — either via multiple wires and an indication device, or by using a time-based coding scheme like Morse code. Extrapolations of this technology to telegrams and telex merely return to the written word as the coding method, and the physical transport is replaced with an electrical equivalent.

The telephone returns to the spoken word, but uses electrical signals to transmit an audio signal across distances. Facsimile reuses the technology of the telephone to provide the transmission of pictures, and although this does not seem significantly different to a telegram or telex message to users of languages which use the small number of characters like the Roman or Cyrillic letter forms, for users of pictorial languages with large numbers of characters like Chinese or Japanese the facsimile is a very different communications device indeed. The videophone combines a telephone with, typically, a video picture of the two participants — although it is sometimes also possible to replace the video with the ability to share a common view of a picture or document.

All of the spoken or visual systems mentioned so far allow communication between groupings of people — either 'one-to-many', or 'many-to-many'. In contrast, the electrical systems have been 'one-to-one', where two participants can communicate either by coded words or by audio. Extending this 'communication between two points' to multiple points uses the concept of a gathering together of people to talk — a conference. Teleconferencing is the process of linking together several participants in a multi-point conversation using telecommunications, and it can use audio and video as well as additional data (text or pictures). Just as the telephone allows people to talk even though they are geographically separated, so teleconferencing allows a conference to be held when the participants are in different locations.

To arrive at a desktop teleconferencing system, the way that the telephone and the desktop work needs to be considered, and then the standards infrastructure that is required to make a broadly applicable product, which exploits the merging of telephone and computer, needs to be looked at.

2. Telephony — two-way interaction

The telephone appears such a natural device for providing conversational interaction that it is now difficult to envisage a world without it. But the original intention of the telephone was to provide access to audio signals at a distance — which is where the name comes from. The word 'telephone' is made up of two Greek parts — *tele* meaning 'remote' and *phone* meaning 'voice'. Comparisons between the radio and the telephone as a way of listening to music or speech appear unlikely in a modern context, but the use of the telephone to carry out conversations which are disjointed by distance is a serendipitous result of the technology, not its original intended usage.

Although it tends to be used mainly for person-to-person communication, it is possible to use the telephone to connect several people together — this is called audio-conferencing. Compared to a face-to-face situation where several people are gathered around a table, audio-conferencing provides no visual information about who is speaking, who is shuffling, or who keeps glancing at their watch. Telephone calls to more than one other person, with or without video, tend to be used mostly for business purposes. Adding video pictures to the telephone can provide some of the information that is lost when using just audio connections. Despite this, the videophone has only slowly moved from speculative fiction to real-world usage, whereas the telephone is almost ubiquitous.

3. The desktop — one-way retrieval

The desktop is the word used for the graphical user interface (GUI) that is presented on the screen of many personal computers. It is called 'the desktop' because it uses some of the elements of a real desktop to provide a metaphor for controlling the computer. The wastebasket or trash-can is perhaps the most obvious example — to throw something away you put it on top of a small picture or 'icon' that represents the wastebasket, and the item is then deleted by the computer. Documents are represented as icons which have the appearance of documents — although they are still stored as data files inside the computer. Storage locations inside the computer are represented by icons which appear either as pictures of the storage device itself (a floppy or hard disk, perhaps) or as a folder to indicate several documents filed in one place. The contents of an icon or folder can be displayed in an on-screen 'window' area — the equivalent of opening up the folder and looking at a catalogue or directory of the contents. To move documents from one place in the computer to another, they are dragged from one icon, folder or window to another.

Because many of the desks found in IT-aware companies are equipped with a telephone and a networked computer, the desktop is a familiar interface to many business users. Unlike the telephone, the desktop has only been in common usage for a few years, and so its use is still developing, although by adopting the metaphor of a real desktop it carries with it a number of associative characteristics. The difference between resources which are local to the computer, and resources which are provided by a local area network (LAN), is hidden from the user — they appear as icons or windows, and can be manipulated in exactly the same way. This is being extended to wider network resources like the Internet, where future operating systems and GUIs will provide effectively 'transparent' access to the WWW.

The main focus of the desktop is one-way — often just the retrieval or publishing of items of information. Just as sending a letter necessarily involves a time-delay while it is in transit, processed and a response sent back, so desktop systems tend to have corresponding delay mechanisms. E-mail merely shortens the time in which a reply may be received, but provides no guarantee that it will be any faster than conventional mail. The Internet provides access to huge quantities of information, but the inherent interaction is normally with a remote database rather than a remote person, and so is characterised by the download of large quantities of text and pictorial information, while the upload of control signals normally occupies significantly less bandwidth.

Business users who have a desk equipped with a telephone and networked computer are also likely to be users of conferencing. At the moment, conferencing is synonymous with audio or video conferencing, and, while the desktop may be involved in the set-up of the conference, it is rarely used interactively other than as a way of displaying the video picture. When collaborative tools are provided, the use of interactive working with another person is unfamiliar in the context of a desktop, and so is under-utilised. The desktop metaphor needs to be extended so that it also becomes associated with interactive bi-directional usage instead of uni-directional retrieval or publishing activities. This will involve a change in the public perception of the telephone and the computer, but it also requires an amalgamation of their different standards and methods of working. Computer/telephony integration (CTI) is one aspect of this merging process [1].

4. Standards

The convergence of the once separate worlds of computing, telecommunications and the media is now an accepted part of the fabric of society. The increasing use of digital technology has acted as the catalyst in this ongoing process, and it brings with it a dependence on the use of defined standards so that successful interworking can happen. Integrating the many layers of standards, protocols and interfaces together into a coherent form involves slotting together different paradigms and metaphors.

Some of the organisations that are working on stabilising this convergence include DAVIC, the ITU, and the IMTC. The convergence itself is driven by both the standards and the technology. In the case of DAVIC, it is the delivery of multimedia content to end users that is the focus, whereas the IMTC works towards facilitating multimedia teleconferencing using standards like T.120 (see Fig 1) and the H.32x series.

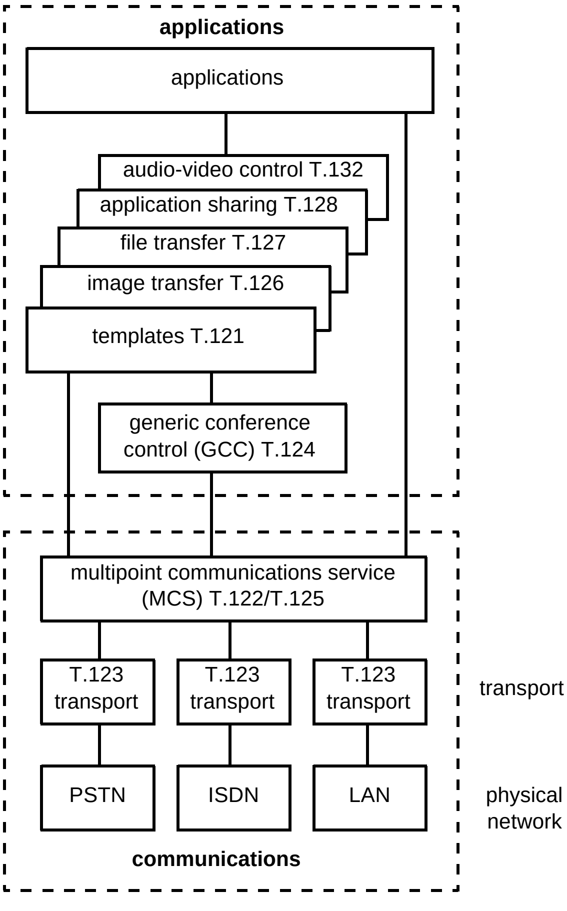

Fig 1 An overview of T.120.

One of the underlying enabling standards for conferencing on the desktop is T.120 [2]. The aim of the ITU T.120 Recommendation is to ease the development of a worldwide dial-up multimedia data conferencing service [3]. It defines a series of open data conferencing standards that can support integrated communications between users, regardless of the computer platform or network technology that is being used. The intention is to make a 'multimedia' call as simple to carry out as making a telephone call.

The T.120 suite of recommendations describe a series of communications and applications protocols and services that can provide support for multi-point data communications [4]. The facilities and abstraction that it provides should make it easier to develop collaborative applications which use diverse types of network connections. Some possibilities include desktop data conferencing, several users sharing a single document on one application, and multi-player gaming. The functionality provided means that it is possible for a single conference to incorporate calls made using many different types of data transport mechanism and end-point capabilities — and yet the conference can be controlled as a single unit. It is possible for the user terminals to exchange information about their capabilities, and then to take these into account in the management of the conference.

T.120 tries to resolve complex technological issues in a way that is acceptable to both the computing and telecommunications industries. As there are no platform dependencies, no major limits on the network topology (star, chains, cascades, etc), it offers easy scalability, extendability, and it will co-exist with other ITU standards. It is intended to be the key building block for conferencing — a unifying element.

The T.120 standards suite is designed as a series of layers, with protocols and service definitions linking them together. The lower layers provide a generic mechanism which allows applications access to multipoint communications services. The higher levels define some specific protocols for use by applications to ensure baseline compatibility — 'whiteboards' or file transfer — although it is possible for an application to utilise other non-standard protocols.

At the lowest level is the transport protocol, known as T.123. This interfaces the T.120 data packets, known as protocol data units or PDUs, to the physical networks: PSTN, ISDN, or LANs (TCP/IP or IPX). This provides a uniform guaranteed delivery of data — and can be seen as being the provider of network interfacing. Above these is the Multipoint Communication Service (MCS), described in T.122 and T.125. The MCS is the heart of the T.120 architecture — it connects together multiple channels of information to form what is known as a domain, roughly equivalent to a 'conference'. The MCS controls the information flows between the applications running on the participant computers, and can provide broadcast and private communications. It thus enables the delivery and synchronisation of data across a multiple set of end-points — a multipoint conference.

The Generic Conference Control (GCC), T.124, sits above the MCS, and turns the generic connections of the MCS into a support mechanism for conferencing. The GCC maintains a database of the conferences in progress, and so serves as the common point for initiating, managing and controlling conferences. Applications register with the GCC, and it controls the assignment of resources from the MCS and below. The GCC is thus providing the conference administration interface, and it co-ordinates the use of the dynamic data resources below it.

The layers above the GCC can communicate either directly to the GCC and MCS, or can use a defined resource management template which is defined in T.121, the Generic Application Template (GAT). The GAT acts as a standard glue between applications and the lower levels, and so helps to ensure consistency and robustness in their interactions. It provides several services such as an enrolment service for applications, initiating applications on other nodes in the conference, and attaching an application to a domain in the MCS. By providing these facilities the GAT aims to reduce interoperability problems. T.126 is a still image protocol which is intended to facilitate a shared pictorial document, commonly called a ‘whiteboard’. It provides virtual workspaces which are shared between the participants, and which can be manipulated by any node. It defines how the images are shared and managed, as well as allowing interactive pointing and annotation by the use of distinct ‘planes’ of view. T.127 is the file transfer protocol, and enables efficient transmission of files to any or all of the nodes in a conference. With advanced implementations, this allows remote information retrieval from archives to be achieved.

BT has been a major contributor to the T.120 standard, and the knowledge and expertise of the specialists is a valuable asset when designing a conferencing system. The remainder of this paper will show how conferencing based on these underlying standards can provide an integrated solution to communicating between several people in a rich environment.

By hooking into the shared information in the conference, Databeam’s ‘neT.120’ server provides a simple way of taking the ‘whiteboard’ content of a T.120-based conference and publishing it on the Internet as a WWW page. It is also possible to use T.120 to provide even closer coupling between applications running on terminals in the conference, by using application sharing. This enables a computer application program on one computer in the conference, to be viewed at all of the other end-point terminals by managing the control of the application, and the distribution of its graphical output. Arbitration of the control devices, typically the mouse and keyboard, enables the application to be controlled from a different end-point to the one where it is located.

For real-time media information streams like video or audio data, as well as management functions like control and indication, then a series of proposed extensions to the basic T.120 recommendations are nearly complete. These are known as the T.130 Audio-Video Control (AVC) standards, and they deal with the way that T.120 handles these types of media streams. The component standards in the T.130 series provide a network-independent control protocol for real-time audio, data and video streams. T.130 uses T.120 data communications services. The movement of data across diverse network boundaries is co-ordinated and managed by T.130, so that end-point terminals with differing capabilities can co-exist in the same conference. When real-time data streams are being used, the quality of service (QoS) and interactions with existing control protocols become important. T.130 enables requests and measurements to be made of the QoS, and has links with the ITU ‘H’ series of control standards — H.242/3 and H.245.

The T.130 suite enables the manipulation of real-time data streams within an established conference topology, but it does not define the mechanisms which are required to set up that topology. The terminal need not implement all of the protocols referenced in T.130 in order to participate in the audio or video parts of a conference, but in this case the control functionality may be limited. The range of control services that is provided includes:

- video switching,
- video processing and transcoding,
- audio mixing and muting,
- control of the ‘floor’,
- identification and selection of multimedia sources,
- channel management in real time.

BT is also leading the development of this new standard, and the final result will be a comprehensive tool-kit of standards and protocols that will facilitate feature-rich teleconferencing.

5. Conferencing

The generic word for conferencing which uses telecommunications is teleconferencing — literally, remote conferencing. There are several different types of teleconferencing, and it is normal to describe them by the major method used to deliver information, although audio is normally assumed to be also present in videoconferencing. A video picture on its own would have limited usefulness compared to an ‘audio-and-video’conference!

The concept of a teleconference is to provide to the participants the same types of facility that they would expect to find at a real meeting or conference — within the restrictions of the media being used. So a desktop conference is limited in the way that it can provide a simulation of a room environment, whereas a videoconferencing suite already has the environment. Immersive virtual reality techniques may alter these divisions in the future, and provide more of the ‘sense of being there’ that is promised by telepresence. Current conferencing techniques tend to be constrained by pragmatism — which in this case means making the most of available technology. On the office desktop, this probably means using just the telephone and networked personal computer in their basic form.

5.1 Audioconferencing

Audioconferencing uses the telephone to provide audio communication between the participants. Three-way calls can be set up directly by the user employing network services, but for more participants special bridging hardware is required — and this is provided by conferencing bureaux services like BT’s Conference Call [5]. Gibson et al give more detail on the implementation of this service [6].

5.2 Videoconferencing

Videoconferencing adds video pictures to an audio-conference. There are two main sub-divisions — room-based systems, and desktop systems. Room-based video-conferencing is normally used by more than one participant, and can be either a permanent installation, or mobile equipment can be moved into a room as required. Because the aim of a room-based system is often to simulate a conventional meeting, then the participants normally sit on one side of a table, whilst the video screens, cameras and loudspeakers are placed on the other side. Desktop video-conferencing is usually designed for single users, and is often associated with a personal computer combined with a microphone and loudspeaker which provide the functionality of a telephone, although stand-alone desktop units are also used. Desktop systems can use either the telephone handset, a headset, or a loud-speaking telephone, and are often designed around the metaphor of a telephone with added video pictures, rather than a meeting room.

Although a videoconference provides the pictures of the participants that are missing in an audioconference, the cost and technical complexity of providing a usable environment that involves audio and video can be prohibitive. Desktop videoconferencing tends to provide additional tools to facilitate the transmission of associated material like documents and pictures, but the link between the video picture on the screen, and documents on the computer is often not easy to comprehend.

5.3 Data-conferencing

Data-conferencing typically uses the Internet to provide communications between participants, often using text-only shared ‘chat’ applications for exchanging written information, or shared picture-based ‘whiteboard’ application for annotating diagrams, maps or other pictorial information. Some Internet-based dataconferencing is point-to-point, which restricts the sharing of the data to just those two people. Multipoint dataconferencing enables more than two participants to exchange or share information, and so work collaboratively.

There are many applications for dataconferencing. They include:

- linking sales teams together to share information,
- linking project managers to co-ordinate their schedules and plans,
- enabling marketing teams to work jointly on presentation material,
- enhancing distance learning applications,
- bringing together students from different locations for special lessons or lectures.

5.4 Audiographic conferencing

Audiographic conferencing brings together audio-conferencing with data-conferencing, and so provides the familiar interaction of a telephone combined with the visual text or pictorial information that is often the focus of a technical or business meeting (see Fig 2).

This produces a dichotomy of user interface — the audio from the telephone is familiar, although loud-speaking telephones perhaps less so, while the presentation of the visual information is probably more unusual, especially in the context of the normal unidirectional retrieval-based use of a computer screen.

One of the major challenges of audiographic conferencing is thus to present the linking between the audio and the data parts of the conference in a way that makes sense to the user. The telephone is strongly associated with audio use, while the desktop uses the metaphor of a desk, and there are no obvious linkages between them. Solving this problem is the key to making desktop conferencing a success.

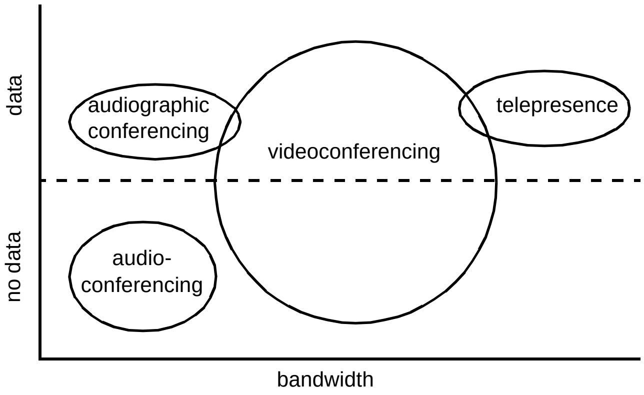

Fig 2 Audiographic conferencing extends audioconferencing by adding in data, while providing a coherent upgrade path towards integrating video and ultimately providing telepresence features into conferencing environments.

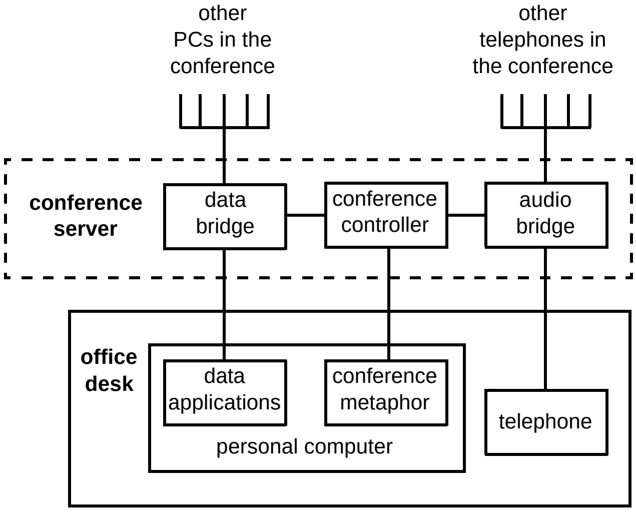

Fig 3 Desktop conferencing. The ‘conference metaphor’ is the on-screen link between the telephone, the participants and the data sharing applications.

5.5 Desktop conferencing

The term ‘desktop conferencing’ is used here to emphasise the use of the computer desktop in the role of an interactive, conversational tool. If the normally retrieval-based desktop is to be used by participants in a conference, then there needs to be a logical linkage between the audio and the visual elements. This can be observed in audioconferences where it is not possible to know how many participants are present, unless they identify themselves individually. If a new participant joins the conference and does not speak, then it is possible for their presence to go unnoticed. Also, the exchange of supporting visual information in such a conference has to be carried out either beforehand, by post or fax, or afterwards — which may well trigger another meeting because the material does not match with a participant’s impression of its content.

A desktop conference needs to bring together the telephone and the computer screen area into one integrated working environment. The actual media forms used merely provide extra information, and do not change this basic requirement. So a desktop conferencing system design must be able to cope with a range of interconnection possibilities — from a simple dataconference, via audioconferencing and audiographic conferencing, to videoconferencing. The provision of a single consistent user environment is the catalyst that makes desktop conferencing a serious communications tool (see Fig 3).

5.6 Advantages of teleconferencing

The key advantages to teleconferencing are summarised below:

- cost-effective — can reduce travel costs, and can have low hardware and software costs,
- consistent — all participants receive the same information at the same time,
- enabling — meetings can take place regardless of distance, and can be set up without worries about long journeys,
- exciting — people can exchange ideas from anywhere in the world, to anywhere else,
- faster — travel time can be reduced,
- involving — interaction is immediate, instead of delayed by post, fax or e-mail,
- ubiquity — people can participate from geographically diverse sites by using a variety of transport mechanisms.

6. Environments

Trying to determine the exact requirements of people who wish to interact in a conference environment is not trivial. Audioconferencing can provide audio links to replace the spoken word, but does not provide any visual information about the participants. Videoconferencing enables the participants to see each other and thus receive the non-verbal ‘body language’ signals that accompany speaking — people still nod their heads and gesticulate with their hands when they are using the telephone. Data-conferencing provides shared access to information that is required in the conference — text documents, diagrams or numbers.

However, the actual working environment of a conference is often very different from the real-world situation. An audio-conference, where people gather around a loud-speaking telephone, does not immediately provide information about who else is in the conference —

'disembodied' voices are all that can be heard. Text documents can be discussed via data-conferencing, which enables several people to contribute to a shared document or diagram, but the separation between the contributor and a real person is similar to audioconferencing — having a name, associated with some text that is appearing on the screen, provides only the bare minimum of identification. Videoconferencing provides the missing visual information about the participants in an audioconference, but after the initial introductions, the main focus of many business meetings is not the participants, but the topic that is being discussed.

Other aspects of a real-world conference that are easily overlooked when a teleconferencing environment is being designed include the set-up of the meeting, and any subsequent follow-up after the meeting. For all types of meeting, the initial pre-meeting action is normally to circulate an agenda to all of the participants. This is normally dealt with by the organiser of the meeting, and follows the booking of the required resources: time, date, room for 'n' people, overhead projector, whiteboard, video recorder, television, etc. For audioconferences in particular, text or pictorial information that is needed in the meeting is often sent by fax or post to all the participants, so that they will have the information in front of them when the meeting starts.

Audiographic conferencing combines the audio-conference with data — the additional text or pictorial information that often forms the focus of the discussion in the meeting. The dataconference can provide several facilities to enable this to happen:

- by allowing documents to be transferred 'in the background' between participants during the meeting, there is no need to send faxes or post printed documents,
- shared 'live' whiteboards allow diagrams or pictorial information to be annotated, revised and discussed *in situ*, as part of the ongoing meeting,
- text documents or spreadsheets can be collaboratively worked on by 'sharing' a suitable application, so that, for example, the exact wording of a paragraph can be determined by consensus, or revisions to the draft of a paper can be agreed by the co-authors.

However, audiographic conferencing can also bring additional facilities to the teleconferencing environment because it is part of the computer desktop. The initial booking of the meeting or conference requires the organiser to determine a suitable date, and this may involve contacting prospective participants in order to determine their availability. This could utilise the telephone for person-to-person communication, but could alternatively be an examination of their desktop calendar/diary software. The organiser then needs to book or reserve the conference, which can be achieved by a number of methods — human interaction with a booking agent, an interactive voice-prompt system, or by using the desktop, i.e. e-mail, or a WWW form page.

The pre-meeting agenda and any background documents may well be transmitted by using e-mail, and this offers the possibility that it can be integrated into the audiographic conference booking system. The e-mail messages that give details of the agenda can also be used to provide the details of the audiographic conference set-up — so that joining is eased for the participants.

During the meeting, audiographic conferencing provides control facilities which enable participants to mute their microphones or loud-speakers, and 'chair' control features which allow one participant to take control over the conference, thereby requiring people to 'request to speak' and then choosing who can speak. This is made possible because of the linking between the desktop dataconference and the audio bridging equipment which controls the telephone lines.

At the end of the meeting, or perhaps immediately afterwards, the organiser may need to book a follow-up meeting, which might be for a week or a month's time, using the same facilities and the same participants — or a slight variation. Being able to edit the booking from the desktop is more efficient and ensures that the participants have continuity between the meetings. Because of the integration of e-mail into the conference booking, it can be augmented with mailing list news and FAQ distribution for registered users.

7. Passepartout

Passepartout brings together the experience of telephony, audioconferencing, T.120 expertise, and knowledge of data networks into a single unifying tool for teleconferencing. Passepartout is an audiographic conferencing system, developed at BT Laboratories in collaboration with the BT Conference Call team, which is designed to ultimately bring the whole of the teleconferencing environment to the desktop. It integrates many of the wider elements associated with conferencing, which have been described above, into one coherent system, while also providing the strong linking between the audio and data parts of the conference. In the terminology of Fig 3, Passepartout is the 'conference metaphor'. The intention behind Passepartout is to provide a complete tool for enabling productive meetings or conferences with geographically separated participants.

The initial meeting or conference set-up is accomplished by using the e-mail message containing the agenda. This also has the details of the teleconference which are required to start the dataconference part of the audiographic conference. Once the data link is established, the participant's telephone will then ring, and they are connected into the conference. The initial focus of the interaction is thus the desktop e-mail message, followed by the opening of the Passepartout window and the ringing of the telephone. This establishes the desktop as the centre of the participant's attention.

The linking between the audio, the data, the control, and the participants is achieved by using a window which contains a representation of the meeting or conference (Fig 4). In its simplest form, this is a room with a table in it, around which icons appear to indicate the participants. Using this metaphor of a room, or perhaps a video-conference, is a very effective way of providing intuitive information about what is happening in the conference. When a new participant joins the conference, a new icon appears. Small glyphs, associated with each participant icon, indicate when they can talk, listen, or when they are requesting to speak. Highlighting of an icon can be used to indicate that they have taken over chairman control, or that they currently 'have the floor'.

Data facilities like file transfer, whiteboarding, and application sharing, are also available from the Passepartout window, so that it is the centre of the participant's interactions with the conference. The telephone is only used to provide audio.

To summarise, the Passepartout audiographic conferencing trial system allows users to:

- monitor and control the conference (see who is in the conference, chairman control, etc),
- use the power of the PC to discuss much more complex issues than in a traditional audio conference — tools are provided for remote presentations, sharing documents and spread sheets, etc, the focus for accessing all of these tools being the Passepartout window toolbar and menus,
- easily join the conference — a WWW page could be used for booking the conference, although a telephone booking service could also be used; Passepartout would then show new icons as participants join the conference, with the participant's telephone ringing and then using a voice prompt when answered to indicate that they had joined the conference; as the integration of the desktop metaphor into conferencing proceeds, there are ways to improve this further — with, for example, shortcuts on the desktop,

Fig 4 An early development version of Passepartout. This shows the on-screen window, with four people in a conference.

- use their intranet (normal LAN connection) in the office and a modem for dial-up access when away from the desk — T.120 is flexible in its use of networking, and can operate over PSTN telephone lines with a modem, ISDN lines, LAN networks, the Internet, or a modem in a GSM mobile phone.

Passepartout does not require additional hardware in the computer. An office desk with a telephone and a computer (networked or with a modem and an additional telephone line) are the only requirements for audiographic conferencing apart from the software. Because the audio is carried by the telephone, the computer does not need processing power to be used for coding and decoding audio signals carried over the data links, and so a modestly specified computer will give perfectly acceptable performance.

Passepartout exploits BT's comprehensive knowledge of T.120, gained during the years of developing the standard, and so gives a tangible expression to the vision and forethought of the team that have contributed to the standards suite. It represents one aspect of the T.120 view of teleconferencing turned into a desktop controller — in this version enabling audiographic conferencing to happen in a convenient and controllable way.

Future enhancements to the Passepartout concept will continue to build on this in-depth knowledge of the standards, and may well start to incorporate the T.130 standards suite as the audiographic conferencing metaphor is extended to include additional types of multimedia information.

8. Conclusions

Interactive communication using the telephone provides an excellent starting point for a number of enhancements that make the most of emerging technologies. Video-conferencing adds a strong visual element so that the participants can be identified and their body language observed. However, audiographic, i.e. audio and data, conferencing links together the audio and data to provide information about who is in the conference without the overhead of video technology. It also allows essential text or pictorial information to be collaboratively worked on when it is required in the meeting — there is no need to fax or e-mail the documentation to all the participants before (or after) the meeting.

The convergence of the underlying technologies and standards is bringing together the fields of conferencing, multimedia, and telephony into a framework which makes the conversational use of the desktop possible. The technology is emerging, and the enabler to the success of teleconferencing on the desktop is the bringing together of a complete system under one unifying metaphor. By easing and simplifying the processes involved in booking, joining, participating, controlling and following up after a meeting or conference, such a comprehensive desktop conferencing system produces a powerful 'on-screen' focus for interactive communication.

The telephone is likely to remain the natural route for conversing with remotely located people, but making the most of a telephone linked to a desktop conferencing system will enable a media-rich conversation instead. Future desktop conferencing systems will see the further integration of data conferencing with directories, calendars, schedulers and other personal productivity and time-management tools. In an increasingly multimedia-based world, desktop conferencing is in an ideal position to enhance and improve business communications.

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